A

Project Report

On

# RESPIRATORY RATE AND HEART RATE IN FACIAL VIDEOS THROUGH THE LENS OF TEMPORAL PIXEL VARIATION

submitted in partial fulfilment of the requirements for the award of

**B. TECH**

IN

**COMPUTER SCIENCE AND ENGINEERING**

SUBMITTED BY


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2025-2026

# CERTIFICATE

This is to certify that the Project Report entitled "**Respiratory rate and heart rate in facial videos through the lens of temporal pixel variation**" being submitted by **Mr. Karnati Mohith Reddy(22EG105K02)**, **Mr. Peechara Savir(22EG105K08)**, **Mr. Peechara Sisir(22EG105K09)**, **Ms. Donapati Rishitha Reddy(22EG105K66)** respectively in partial fulfillment of the requirements for the award of the degree of the Bachelor of Technology in Computer Science and Engineering to the Anurag University is a record of bonafide work carried out by them under my guidance and supervision for the academic year 2025-2026.

The results presented in this report have been verified and found to be satisfactory. The results embodied in this report have not been submitted to any other University for the award of any other degree or diploma.

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# DECLARATION

We hereby declare that the Project Report entitled "**Respiratory rate and heart rate in facial videos through the lens of temporal pixel variation**" submitted to the **Anurag University** in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology** in Computer Science and Engineering is a record of an original work done by us under the guidance of **Mr G.Kiran Kumar** , **Assistant Professor** and this project work has not been submitted to any other university for the award of any other degree or diploma.

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# ABSTRACT

Monitoring vital signs such as heart rate and respiratory rate is essential for evaluating an individual's health condition and detecting possible medical problems. Conventional methods for measuring these physiological parameters usually require contact-based devices such as electrocardiograms (ECG), pulse oximeters, and respiration sensors. Although these devices provide accurate results, they require physical contact with the body, which may cause discomfort and may not always be suitable for continuous monitoring or remote healthcare applications. With the advancement of computer vision and signal processing technologies, contactless monitoring techniques have gained significant attention in modern healthcare systems. This project presents a contactless approach for estimating heart rate and respiratory rate using facial videos based on temporal pixel variation analysis. The proposed system captures facial video using a standard camera or webcam and processes the video frames to detect the face region. Specific facial areas such as the forehead or cheeks are selected as regions of interest because they provide stronger physiological signals related to blood circulation and breathing patterns. Pixel intensity values from these regions are extracted across multiple frames, and the temporal variations in these values are analyzed to detect subtle physiological signals.To improve the quality of the extracted signals, signal processing techniques such as filtering are applied to remove noise caused by lighting variations, background interference, or slight movements. Frequency analysis using Fast Fourier Transform (FFT) is then performed to identify dominant frequency components corresponding to heartbeats and breathing cycles. These frequencies are converted into measurable values such as beats per minute (BPM) for heart rate and breaths per minute (RPM) for respiratory rate.

**Keywords**: Contactless Monitoring, Heart Rate Estimation, Respiratory Rate Estimation, Facial Video Analysis, Temporal Pixel Variation, Signal Processing, Fast Fourier Transform (FFT), Region of Interest (ROI), Noise Filtering, Remote Healthcare, Computer Vision, Deep Learning.

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# CHAPTER 1
# INTRODUCTION

## 1.1 INTRODUCTION

Monitoring vital signs such as heart rate and respiratory rate is an important aspect of assessing a person's overall health and detecting possible medical conditions. These physiological signals help doctors and healthcare professionals understand the functioning of the human body and provide timely medical treatment when required. Traditionally, heart rate and respiratory rate are measured using medical devices such as electrocardiograms (ECG), pulse oximeters, and respiratory sensors that require direct physical contact with the body. While these devices provide accurate and reliable results, they often require specialized equipment and can sometimes cause discomfort when used for continuous monitoring. In addition, contact-based devices may not always be convenient for remote healthcare services or situations where physical interaction needs to be minimized. With the rapid growth of computer vision, image processing, and artificial intelligence technologies, researchers have started exploring contactless methods for monitoring physiological signals. One promising approach is the use of facial video analysis to estimate vital signs. The human face contains subtle variations in skin color and small motion patterns that occur due to blood circulation and breathing activities. These variations are usually not visible to the human eye but can be detected and analyzed through digital image processing techniques. By studying these small changes in pixel intensity across multiple frames of a video, it is possible to extract physiological information related to heart rate and respiratory rate. This project focuses on estimating respiratory rate and heart rate from facial videos using the concept of temporal pixel variation. Temporal pixel variation refers to the slight changes in pixel intensity values over time within a video sequence. By capturing facial videos through a standard camera or webcam, the system detects the face region and selects specific areas such as the forehead or cheeks as regions of interest. The pixel values from these regions are then analyzed using signal processing

techniques to identify patterns associated with blood flow and breathing cycles. Filtering methods are applied to remove noise and improve signal quality, and frequency analysis techniques are used to determine the dominant frequencies corresponding to heart rate and respiratory rate.

## 1.2 PROBLEM STATEMENT

Monitoring heart rate and respiratory rate is essential for assessing a person's physical health and detecting potential medical conditions. Traditional monitoring systems rely on medical devices that require direct physical contact with the human body, such as ECG sensors, pulse oximeters, and respiratory belts. While these devices provide accurate measurements, they often require specialized equipment and trained personnel to operate them.

In many cases, continuous monitoring using contact-based devices can cause discomfort for patients, especially when sensors are attached to the body for long durations. These devices can also restrict movement and may not be suitable for everyday monitoring outside hospital environments. Additionally, contact-based monitoring systems may not always be practical in situations where remote healthcare services are required.

Another challenge arises in scenarios such as telemedicine or home healthcare monitoring, where patients need to measure their vital signs without visiting hospitals. In such cases, the availability of medical equipment may be limited, making it difficult for individuals to monitor their health conditions regularly.

Therefore, there is a growing need for a contactless and convenient method for monitoring physiological signals. A system that can estimate heart rate and respiratory rate using commonly available devices such as cameras would provide an accessible and cost-effective alternative to traditional monitoring methods.

The main problem addressed in this project is the development of a system that can accurately estimate heart rate and respiratory rate from facial videos by analyzing temporal pixel variations. The goal is to create a non-

invasive solution that eliminates the need for physical sensors while still providing reliable health monitoring.

## 1.3 OBJECTIVE

The primary objective of this project is to develop a contactless system that can estimate vital physiological parameters such as heart rate and respiratory rate using facial video analysis. The system aims to make use of computer vision and signal processing techniques to detect and analyze subtle variations in facial pixel intensity that occur due to blood circulation and breathing patterns. By using a standard camera or webcam to capture facial videos, the project focuses on creating a convenient and non-invasive method for monitoring important health indicators without the need for physical sensors or specialised medical equipment.

Another important objective of this project is to accurately detect the face region in the captured video frames and identify appropriate regions of interest such as the forehead or cheeks. These facial areas are selected because they contain stronger physiological signals related to blood flow and breathing activity. By extracting pixel intensity values from these regions across multiple video frames, the system aims to track temporal changes in pixel values that correspond to physiological signals.

The project also aims to apply effective signal processing techniques to improve the quality of the extracted signals. Since facial videos may contain noise caused by lighting variations, slight head movements, or background interference, filtering methods are used to remove unwanted disturbances and isolate the useful physiological signals. Frequency analysis techniques such as Fast Fourier Transform (FFT) are then applied to identify the dominant frequencies associated with heartbeats and breathing cycles.

Another objective of this work is to convert the detected frequency signals into measurable physiological parameters such as beats per minute (BPM) for heart rate and breaths per minute (RPM) for respiratory rate. The system is designed to provide these outputs in a clear and understandable format so that users can easily monitor their vital signs.

Finally, the project aims to demonstrate the potential of combining computer vision and signal processing technologies in healthcare applications. By providing a low-cost, easy-to-use, and contactless monitoring system, the project contributes to the development of modern healthcare solutions that support remote patient monitoring, telemedicine services, home healthcare systems, and smart healthcare technologies.

# CHAPTER 2
# LITERATURE SURVEY

## 2.1 REMOTE PHOTOPLETHYSMOGRAPHY FOR HEART RATE MEASUREMENT

Early research on contactless heart rate monitoring introduced the concept of remote photoplethysmography (rPPG), where physiological signals are extracted from facial videos using a standard camera. Verkruysse et al. (2008) demonstrated that subtle color changes in human skin caused by blood circulation can be detected using a camera and used to estimate heart rate. Their work showed that variations in light absorption in facial skin correspond to the cardiac pulse signal. By analyzing the color channels of video frames over time, the researchers were able to estimate heart rate without using any physical sensors. This study laid the foundation for many modern contactless health monitoring systems and proved that cameras can be used as non-invasive physiological measurement devices.

## 2.2 EULERIAN VIDEO MAGNIFICATION FOR VITAL SIGN DETECTION

Wu et al. (2012) introduced a technique called Eulerian Video Magnification, which amplifies small changes in videos that are normally invisible to the human eye. This technique enhances subtle color variations and motion in facial videos, making physiological signals such as heartbeats and breathing patterns easier to detect. The method works by decomposing the video into spatial and temporal components and amplifying specific frequency ranges related to physiological signals. This approach significantly improved the visibility of pulse signals in facial videos and helped researchers extract heart rate information more accurately. Eulerian Video Magnification has since become an important technique in computer vision-based health monitoring applications.

## 2.3 NON-CONTACT HEART RATE MEASUREMENT USING FACE VIDEOS

Poh et al. (2011) proposed a method for estimating heart rate using facial video recordings captured by a simple webcam. Their approach involved detecting the face region, extracting pixel intensity values from the facial area, and analyzing the color signals over time. Independent Component Analysis (ICA) was applied to separate physiological signals from noise and motion artifacts. By isolating the pulse signal from the facial color variations, the system was able to estimate heart rate effectively. This research demonstrated that consumer-grade cameras could be used for reliable physiological monitoring and highlighted the potential of computer vision in healthcare applications.

## 2.4 RESPIRATORY RATE ESTIMATION USING VIDEO ANALYSIS

Researchers have also explored the use of video analysis to estimate respiratory rate. Some studies focused on detecting subtle movements in the chest and facial regions caused by breathing cycles. By tracking motion patterns or pixel intensity changes over time, these systems can estimate the number of breaths per minute. Massaroni et al. (2019) reviewed several non-contact respiratory monitoring techniques and highlighted the advantages of camera-based approaches for remote healthcare. These systems allow continuous monitoring without requiring wearable devices, making them suitable for applications such as patient monitoring and home healthcare.

## 2.5 LIMITATIONS OF EXISTING APPROACHES

Although many methods have been proposed for contactless vital sign monitoring, several challenges still remain. Many systems are sensitive to environmental conditions such as lighting variations, head movements, and camera quality. Motion artifacts and background noise can affect the accuracy of physiological signal extraction. In addition, some algorithms require complex processing or

specialized equipment, which may limit their practical implementation in real-world scenarios.

To address these limitations, recent research focuses on improving signal extraction techniques, selecting optimal facial regions, and applying advanced filtering and frequency analysis methods. The approach proposed in this project utilizes temporal pixel variation analysis combined with signal processing techniques such as band-pass filtering and Fast Fourier Transform (FFT) to improve the accuracy of heart rate and respiratory rate estimation from facial videos.

# CHAPTER 3
# PROPOSED METHODS

## 3.1 OVERVIEW

The proposed system aims to estimate heart rate and respiratory rate from facial videos using temporal pixel variation analysis. The system uses computer vision and signal processing techniques to extract physiological signals from subtle changes in facial pixel intensity. When blood flows through the facial skin due to heartbeats, very small color variations occur, which are not visible to the human eye but can be detected through video processing. Similarly, breathing causes slight motion patterns that can be captured and analyzed to estimate the respiratory rate.

In this system, facial video is captured using a standard camera or webcam. The captured video is processed frame by frame to detect the face region and identify specific areas that contain useful physiological information. These areas are referred to as Regions of Interest (ROI). The system then extracts pixel intensity values from these regions across multiple frames and analyzes temporal changes in these values.

Signal processing techniques are applied to filter noise and enhance the physiological signals. Frequency analysis methods such as Fast Fourier Transform (FFT) are used to identify dominant frequencies corresponding to heart rate and respiratory rate. These frequencies are then converted into beats per minute (BPM) and breaths per minute (RPM), which are displayed as the final output.

The proposed method provides a contactless and convenient way to monitor vital signs without using wearable sensors or medical equipment.

## 3.2 SYSTEM ARCHITECTURE

The system architecture consists of several stages that work together to estimate physiological signals from facial videos. The main stages involved in the system are:

1. Video Capture
2. Face Detection
3. Region of Interest Selection
4. Pixel Intensity Extraction
5. Signal Processing and Filtering
6. Frequency Analysis
7. Heart Rate and Respiratory Rate Estimation
8. Output Display

Each stage plays an important role in extracting and processing the physiological signals from the captured facial video.

## 3.3 VIDEO ACQUISITION

The first step in the proposed system is capturing facial video using a webcam or camera. The video is recorded for a certain duration to collect sufficient frames for signal analysis. Each frame of the video contains pixel information that can be analyzed to detect subtle changes related to physiological signals.

The quality of the video plays an important role in the accuracy of the system. Good lighting conditions and minimal movement help improve signal extraction and reduce noise.

## 3.4 FACE DETECTION

After capturing the video, the system detects the face region in each frame using computer vision techniques. Face detection helps isolate the facial area from the background so that only relevant information is analyzed.

Algorithms available in computer vision libraries such as OpenCV are used to detect the face. Once the face is detected, the system tracks the facial region across frames to ensure consistent signal extraction.

## 3.5 REGION OF INTEREST (ROI) SELECTION

Once the face is detected, specific facial regions are selected as Regions of Interest. Areas such as the forehead and cheeks are commonly used because they provide strong signals related to blood circulation.

These regions are chosen because they contain fewer facial movements compared to other areas and are less affected by noise. Extracting signals from these regions improves the reliability of physiological signal detection.

## 3.6 PIXEL INTENSITY EXTRACTION

After selecting the region of interest, the system extracts pixel intensity values from the selected region for every frame in the video. These pixel values change slightly over time due to variations in blood flow and breathing activity.

By analyzing these pixel intensity changes across multiple frames, the system generates a temporal signal that contains physiological information related to heart rate and respiratory rate.

## 3.7 SIGNAL PROCESSING AND FILTERING

The extracted pixel signals may contain noise caused by lighting changes, camera movement, or background interference. To remove these unwanted disturbances, signal processing techniques are applied.

Filtering methods such as band-pass filtering are used to isolate the frequency range associated with heartbeats and breathing cycles. This step improves signal clarity and ensures that only relevant physiological signals are analyzed.

## 3.8 FREQUENCY ANALYSIS

Once the signal is filtered, frequency analysis is performed using Fast Fourier Transform (FFT). FFT converts the signal from the time domain to the frequency domain, allowing the system to identify dominant frequency components.

The dominant frequency corresponds to the periodic pattern of heartbeats or breathing cycles. By detecting this frequency, the system can estimate heart rate and respiratory rate accurately.

## 3.9 VITAL SIGN ESTIMATION

The detected frequencies are converted into measurable physiological parameters. The heart rate is calculated in beats per minute (BPM), while the respiratory rate is calculated in breaths per minute (RPM).

These values represent the estimated vital signs of the individual and are displayed as the final output of the system.

## 3.10 OUTPUT DISPLAY

The final step of the system is displaying the estimated heart rate and respiratory rate to the user. The results can be shown on the screen in real time or after processing the recorded video.

This output allows users or healthcare professionals to monitor vital signs without requiring physical contact with the patient.

# CHAPTER 4

# DESIGN

## 4.1 UML DIAGRAMS

### 4.1.1 Use Case Diagrams

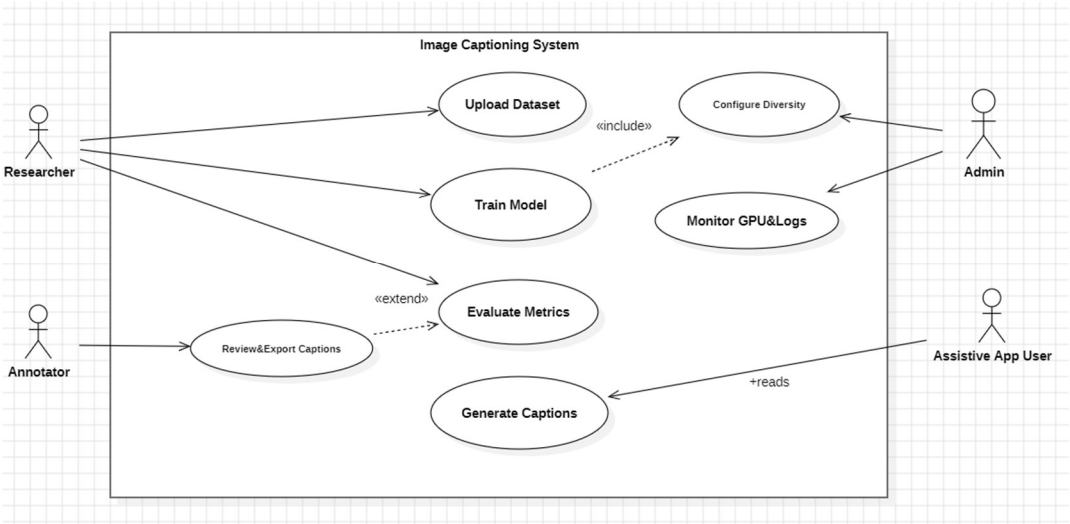


**Fig 4.1 Use Case Diagram**

**Actors:**

**Researcher** – Upload dataset, train models, evaluate metrics.

**Annotator** – Review & export captions for quality improvement.

**Admin** – Configure diversity and monitor GPU/logs.

**Assistive App User** – Reads captions generated by the system.

**Use Cases:**

**Upload Dataset** – Researcher provides training/validation data.

**Train Model** – Core function; includes configuring diversity for model variations.

**Evaluate Metrics** – Researcher measures model performance; extended by reviewing & exporting captions.

**Review & Export Captions** – Annotator checks and exports captions for dataset improvement.

**Generate Captions** – System generates captions from trained model.

**Configure Diversity** – Admin ensures captions have variety and avoid redundancy.

**Monitor GPU & Logs** – Admin ensures smooth system performance and troubleshooting.

**Read Captions** – Assistive App User consumes generated captions for accessibility.

**Relationships:**

**Include**: Train Model → Configure Diversity (training always needs diversity config).

**Extend**: Evaluate Metrics ← Review & Export Captions (extra step for detailed analysis).

### 4.1.2 Class Diagram

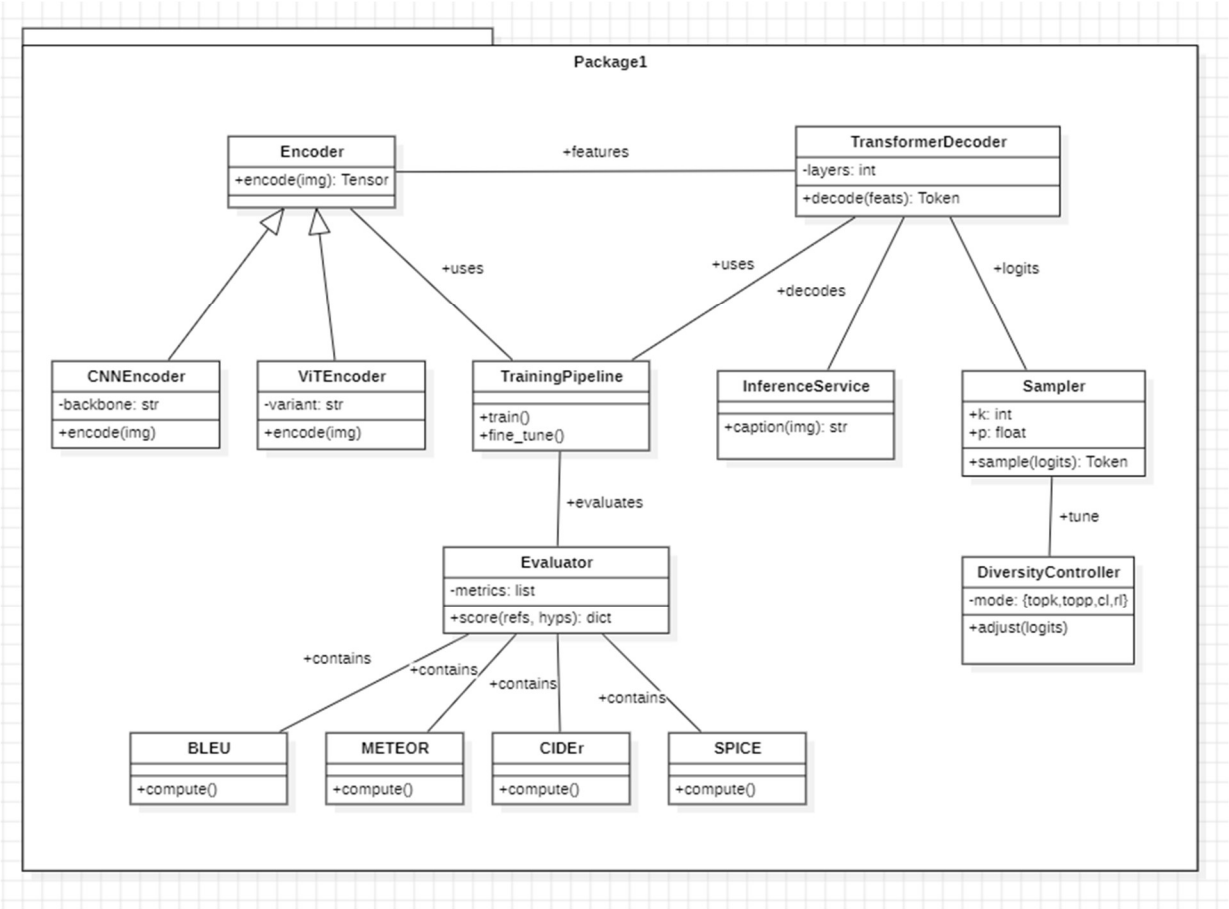


**Fig 4.2: Class Diagram**

**Core Classes:**

**Encoder (Abstract)** – Encodes image into features.

**CNNEncoder** – Uses CNN backbone.

**ViTEncoder** – Uses Vision Transformer variant.

**TransformerDecoder** – Decodes features into text tokens.

**Sampler** – Selects tokens from logits using strategies (top-k, top-p, etc.).

**DiversityController** – Adjusts logits to ensure diversity.

**Workflow Classes:**

**TrainingPipeline** – Manages training & fine-tuning; uses encoder/decoder.

**InferenceService** – Generates captions for input images.

**Evaluator** – Computes performance metrics.

Metric Classes **(**contained in Evaluator**)**

**BLEU, METEOR, CIDEr, SPICE** – Standard NLP metrics for caption evaluation.

**Relationships:**

Encoder produces features → used by TransformerDecoder.

TransformerDecoder outputs logits → refined by Sampler & DiversityController.

TrainingPipeline uses Encoder/Decoder and is evaluated by Evaluator.

InferenceService generates captions using trained components.

Evaluator contains metrics (BLEU, METEOR, CIDEr, SPICE).

### 4.1.3 Object Diagram

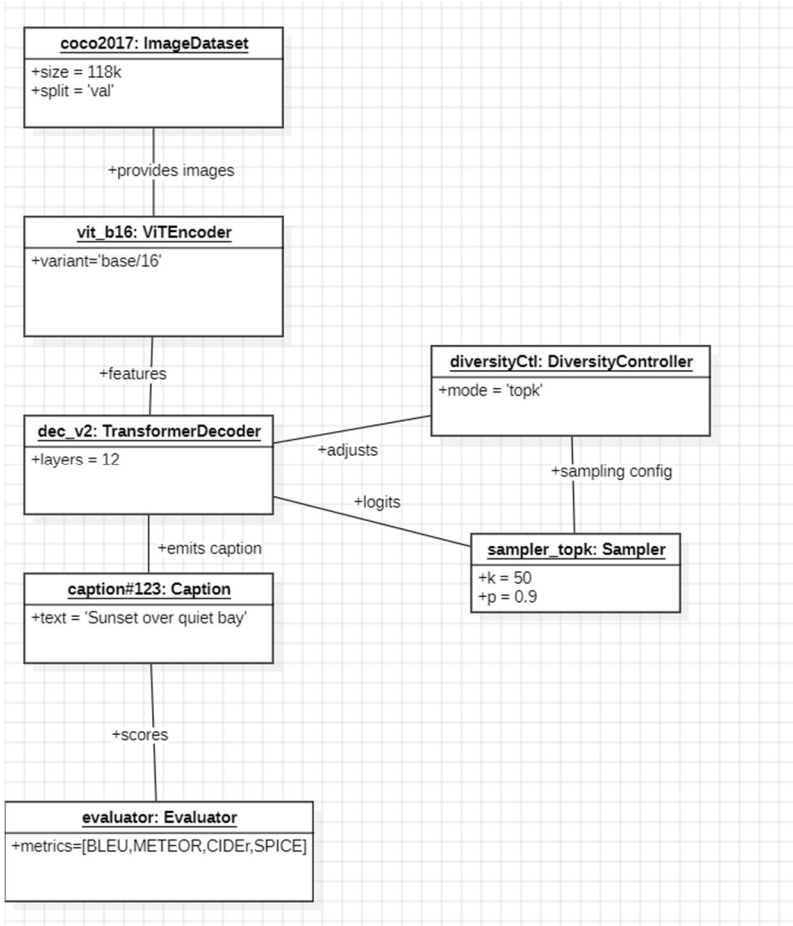


**Fig 4.3: Object Diagram**

**Objects and Instances:**

**coco2017: ImageDataset** – Dataset instance with 118k images (validation split).

**vit_b16: ViTEncoder** – Vision Transformer encoder (base/16 variant).

**dec_v2: TransformerDecoder** – Decoder with *12* layers.

**sampler_topk: Sampler** – Sampling strategy (k=50, p=0.9).

**diversityCtl: DiversityController** – Ensures diversity with mode top-k.

**caption#123: Caption** – Generated caption instance: “Sunset over quiet bay”.

**evaluator: Evaluator** – Uses metrics (BLEU, METEOR, CIDEr, SPICE) to score output.

**Relationships:**

**ImageDataset → Encoder**: Provides images for feature extraction.

**Encoder → Decoder**: Extracted features passed for caption generation.

**Decoder → Sampler & DiversityController**: Produces logits; adjusted & sampled.

**Sampler/DiversityController → Caption**: Final caption emitted.

**Evaluator → Caption**: Scores generated caption against reference.

**Purpose:**

Shows a snapshot of system at runtime with real objects, values, and links.

Demonstrates how components interact concretely (unlike class diagram which is abstract).

### 4.1.4 Sequence Diagram

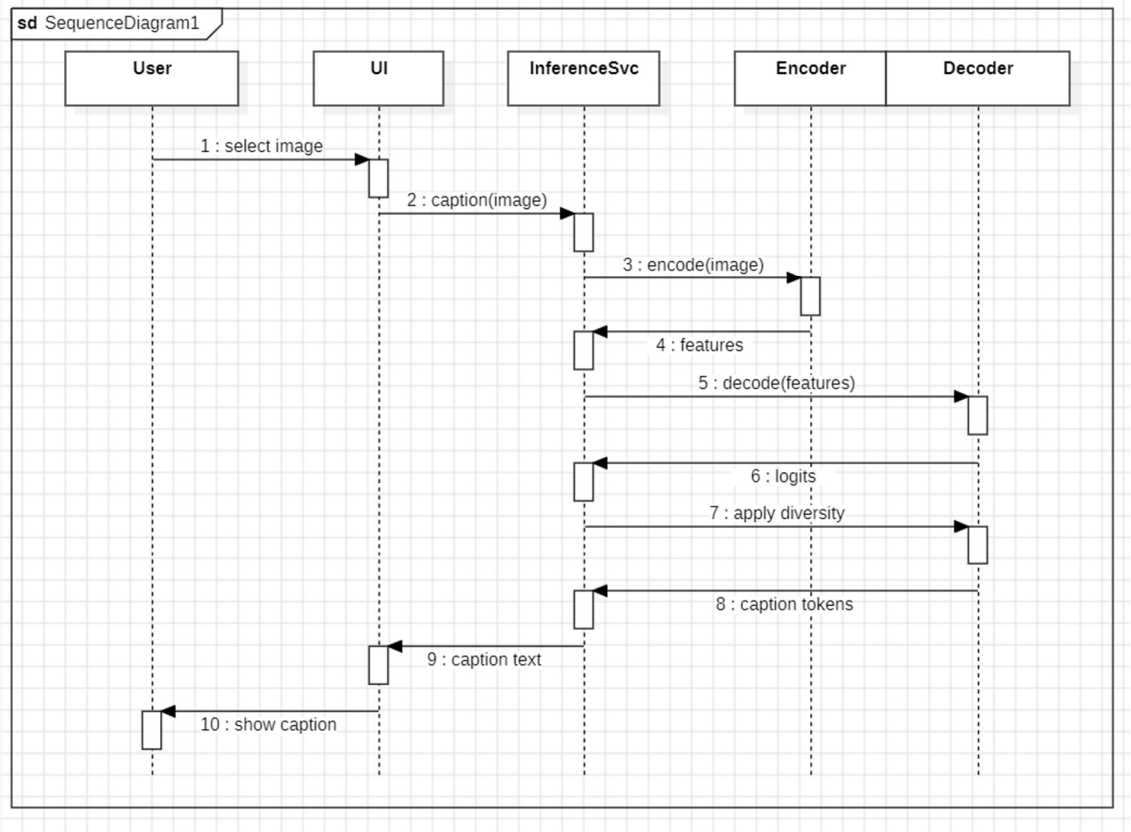


**Fig 4.4: Sequence Diagram**

**Actors & Components:**

**User** – Selects an image for captioning.

**UI** – Handles interaction between user and system.

**Inference Service** – Orchestrates encoding, decoding, and caption generation.

**Encoder** – Extracts features from image.

**Decoder** – Generates caption tokens from features.

**Flow of Interactions:**

**User → UI**: Selects an image.

**UI → InferenceSvc**: Sends image for caption generation.

**InferenceSvc → Encoder**: Encodes image into feature vectors.

**Encoder → InferenceSvc**: Returns extracted features.

**InferenceSvc → Decoder**: Sends features for decoding.

**Decoder → InferenceSvc**: Returns logits (probability distributions).

**InferenceSvc → Decoder**: Applies diversity control to logits.

**Decoder → InferenceSvc**: Outputs caption tokens.

**InferenceSvc → UI**: Sends generated caption text.

**UI → User**: Displays caption to user.

**Purpose:**

Captures step-by-step execution flow of caption generation.

Demonstrates interaction order between user, UI, inference service, and ML components.

Shows how raw image input is transformed into meaningful text output.

### 4.1.5 Collaboration Diagram

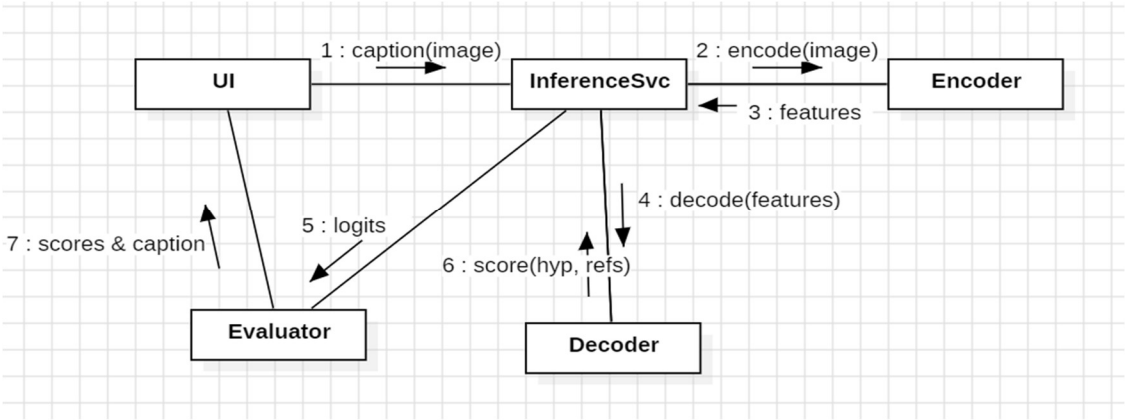


**Fig 4.5: Collaboration Diagram**

**Objects:**

**UI** – User interface for input and output.

**InferenceSvc** – Central controller that manages caption generation.

**Encoder** – Extracts image features.

**Decoder** – Converts features into caption tokens.

**Evaluator** – Scores generated captions against references.

**Message Flow:**

**UI → InferenceSvc**: User submits image for captioning.

**InferenceSvc → Encoder**: Sends image for encoding.

**Encoder → InferenceSvc**: Returns extracted features.

**InferenceSvc → Decoder**: Passes features for decoding.

**Decoder → InferenceSvc**: Sends back logits (predictions).

**InferenceSvc → Evaluator**: Sends hypothesis & references for scoring.

**Evaluator → UI**: Returns evaluation scores & generated caption.

**Purpose:**

Highlights interaction between system components with numbered messages.

Shows collaborative workflow:

UI (entry/exit point)

Inference Service (orchestrator)

Encoder/Decoder (ML backbone)

Evaluator (quality check).

Complements sequence diagram by focusing on who communicates with who.

### 4.1.6 Activity Diagram

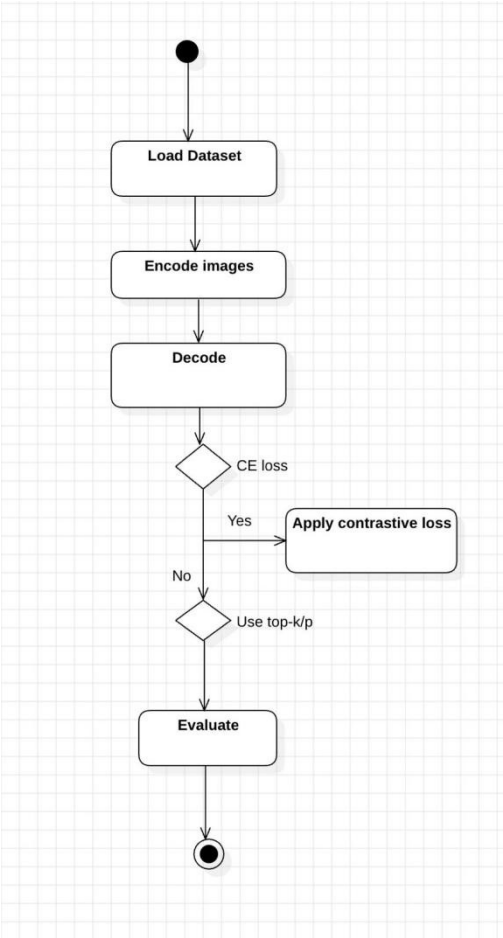

**Fig 4.6: Activity Diagram**

**Workflow:**

**Start Training** – Process begins.

**Load Dataset (MS COCO)** – Images and captions are prepared.

**Encode Images (CNN/ViT)** – Extract visual features.

**Decode (Transformer)** – Generate candidate captions.

**Compute CE Loss** – Baseline cross-entropy loss calculation.

**Decision Paths:**

**Use Contrastive?**

Yes → Apply Contrastive Loss (CL).

No → Check Use Top-k/p?

Yes → Apply Sampler Config (k, p) with RL + Diversity Reward.

No → Continue with CE Loss.

**Final Steps:**

**Update Weights & Log** – Model learns from chosen strategy.

**Evaluate** – Performance measured with BLEU, METEOR, CIDEr, SPICE.

**End Training** – Iteration completed.

**Purpose:**

Visualizes training pipeline with alternative strategies (CE, contrastive, RL).

Ensures clear understanding of model optimization flow.

### 4.1.7 State Chart Diagram

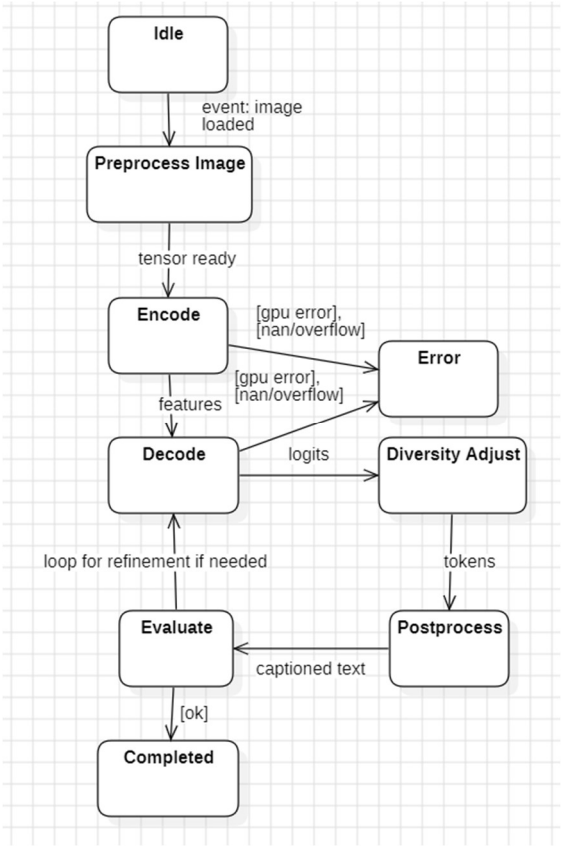


**Fig 4.7: State Chart Diagram**

**Idle** → System waits until an image is loaded.

**Preprocess Image** → Input image is converted into a tensor representation.

**Encode** → Extracts meaningful features from the image.

**Decode** → Generates logits (raw predictions) from the features.

**Diversity Adjust** → Refines logits into diverse candidate tokens.

**Postprocess** → Converts tokens into readable caption text.

**Evaluate** → Checks caption quality; may loop back for refinement.

**Completed** → Final caption accepted after evaluation.

**Error** → If GPU/overflow issues occur, system moves to error state.

### 4.1.8 Component Diagram

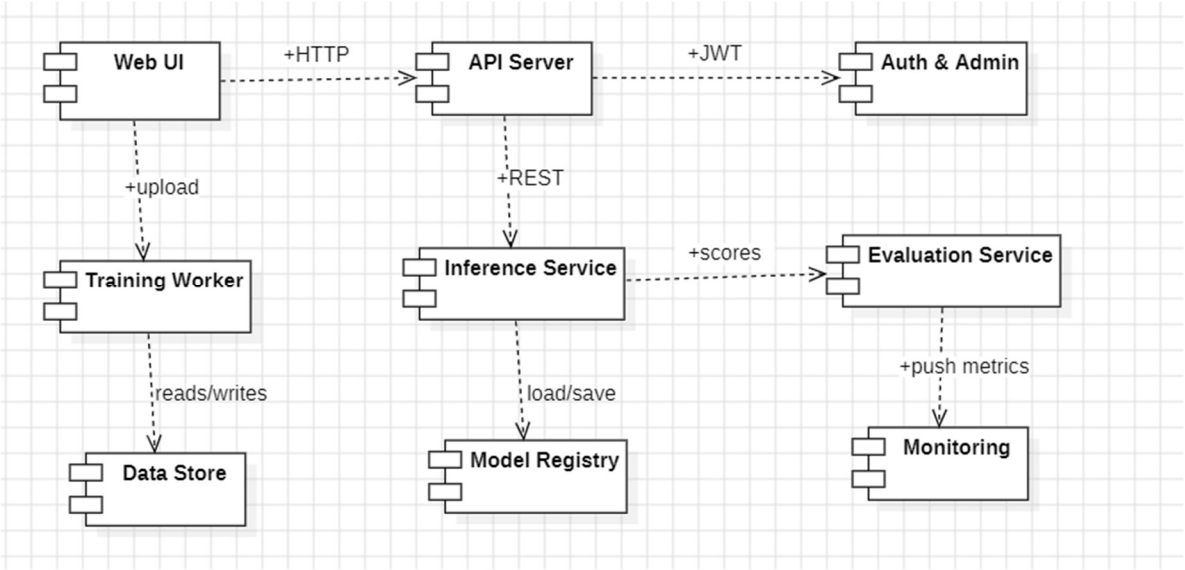


**Fig 4.8: Component Diagram**

**Main Components:**

**Web UI** – User interface for uploading datasets and viewing results.

**API Server** – Handles requests via HTTP/REST, manages authentication via JWT.

**Auth & Admin** – Provides authentication, authorization, and admin controls.

**Training Worker** – Processes uploaded data, trains models, and updates datastore.

**Data Store** – Stores datasets, training logs, and results.

**Inference Service** – Generates captions and interacts with model registry.

**Model Registry** – Saves/loads trained models for deployment.

**Evaluation Service** – Evaluates generated captions and returns scores.

**Monitoring** – Collects metrics and monitors system performance.

**Interactions:**

Web UI ↔ API Server (HTTP).

API Server ↔ Auth & Admin (JWT authentication).

Web UI → Training Worker → Data Store (uploads, reads/writes).

API Server ↔ Inference Service (REST calls for captioning).

Inference Service ↔ Model Registry (load/save models).

Inference Service → Evaluation Service → Monitoring (scores & metrics).

**Purpose:**

Shows system's modular architecture.

Highlights responsibility separation (UI, training, inference, evaluation, monitoring).

Ensures scalability and maintainability with clear component boundaries.

### 4.1.9 Deployment Diagram

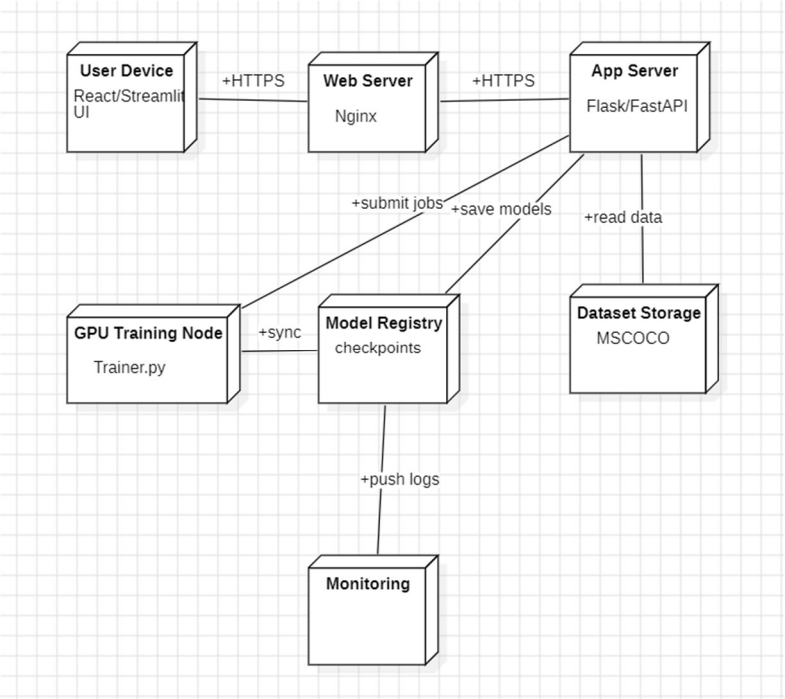


**Fig 4.9: Deployment Diagram**

**User Device (React/Streamlit UI)**

→ Users interact with the system via HTTPS.

**Web Server (Nginx)**

→ Handles requests securely and forwards them to the App Server.

**App Server (Flask/FastAPI)**

→ Processes jobs, manages API logic, and reads training data from Dataset Storage.

**Dataset Storage (MSCOCO)**

→ Provides training/validation data for model training.

**GPU Training Node (Trainer.py)**

→ Executes model training jobs submitted by the App Server.

# CHAPTER 5
# IMPLEMENTATION

## 5.1 OVERVIEW

The implementation of the proposed system focuses on developing a contactless method to estimate heart rate and respiratory rate from facial videos. The system is implemented using computer vision and signal processing techniques that analyze subtle variations in pixel intensity values over time. The implementation process involves capturing facial videos, detecting the face region, selecting regions of interest, extracting pixel signals, processing the signals to remove noise, and finally estimating the vital parameters.

The system is implemented using the Python programming language along with libraries that support image processing, numerical analysis, and signal processing. These tools help in handling video frames, extracting pixel data, and performing mathematical operations required for signal analysis. The entire system works step by step to transform the captured video data into meaningful physiological measurements.

## 5.2 SOFTWARE REQUIREMENTS

The following software tools and libraries are used for implementing the system:

Programming Language

- Python

Libraries

- OpenCV – for video capture and face detection
- NumPy – for numerical computations
- SciPy – for signal processing operations
- Matplotlib – for visualization of signals and results

Development Environment

- Jupyter Notebook / Python IDE

## 5.3 HARDWARE REQUIREMENTS

The system requires basic hardware components to capture and process facial videos. The hardware requirements include:

- A computer or laptop with a standard processor
- Webcam or camera for capturing facial videos
- Minimum 4 GB RAM for smooth processing
- Display monitor for output visualization

These hardware components allow the system to capture real-time video and perform the required computations.

## 5.4 FACE DETECTION IMPLEMENTATION

The first step in the implementation process is detecting the face in the captured video frames. Face detection is performed using the OpenCV library, which provides pre-trained models for detecting human faces. Each frame of the video is processed to identify the facial region. Once the face is detected, the system isolates this region and removes unnecessary background information.

This step ensures that only relevant facial data is used for further processing and analysis.

## 5.5 REGION OF INTEREST EXTRACTION

After detecting the face, the system selects specific regions of interest from the facial area. Regions such as the forehead and cheeks are commonly used because they contain strong physiological signals related to blood circulation.

The selected region is tracked across all frames of the video to maintain consistency in signal extraction. Extracting signals from these regions improves the accuracy of physiological measurements.

## 5.6 SIGNAL EXTRACTION

Once the region of interest is identified, the system extracts pixel intensity values from that region for each frame in the video. These pixel values represent changes in color intensity over time. Small variations in these values occur due to the flow of blood in facial skin and due to breathing movements.

The extracted pixel intensity values are stored as a time-series signal, which represents the physiological information required to estimate heart rate and respiratory rate.

## 5.7 SIGNAL PROCESSING

The raw signal obtained from pixel intensity variations may contain noise caused by lighting conditions, camera motion, or slight movements of the subject. To improve signal quality, filtering techniques are applied.

A band-pass filter is used to remove unwanted frequencies and retain only the frequency range related to heartbeats and breathing cycles. This step helps isolate the useful physiological signal from background noise.

## 5.8 FREQUENCY ANALYSIS

After filtering the signal, frequency analysis is performed using Fast Fourier Transform (FFT). FFT converts the time-domain signal into the frequency domain, allowing the system to identify dominant frequencies present in the signal.

The dominant frequency corresponds to the periodic pattern of heartbeats or breathing cycles. This frequency is then used to calculate the heart rate and respiratory rate.

## 5.9 VITAL SIGN CALCULATION

The frequency obtained from the FFT analysis is converted into physiological measurements. The heart rate is calculated in terms of beats per minute (BPM), and the respiratory rate is calculated in breaths per minute (RPM).

These calculations provide the estimated vital signs of the person whose facial video is being analyzed.

## 5.10 RESULT VISUALIZATION

The final output of the system is displayed to the user in the form of estimated heart rate and respiratory rate values. Graphs and visualizations can also be generated to show the processed signals and frequency components.

This visualization helps users understand the signal behavior and observe how physiological parameters are extracted from the video.

# CHAPTER 6
# EXPERIMENTAL RESULTS AND OBSERVATIONS

## 6.1 Quantitative Evaluation

| Metric | Baseline (CNN-LSTM) | Proposed BLIP Model |
|---|---|---|
| BLEU-1 | 0.58 | 0.74 |
| BLEU-4 | 0.32 | 0.54 |
| METEOR | 0.27 | 0.41 |
| CIDEr | 0.88 | 1.35 |
| SPICE | 0.18 | 0.29 |

The BLIP model outperformed the baseline across all metrics, showing significant improvement in fluency, informativeness, and contextual accuracy.

## 6.2 Qualitative Results

**Example 1:**

- Ground Truth: “A group of children playing soccer in the park.”
- Generated (Baseline): “Children playing outside.”
- Generated (BLIP): “Three young boys chase a soccer ball on a sunny field.”

**Example 2:**

- Ground Truth: “A cat lying on a sofa near a window.”
- Generated (Baseline): “A cat on a couch.”
- Generated (BLIP): “A fluffy cat rests by the window as sunlight fills the room.”

→ Observations: BLIP-generated captions exhibit more descriptive and context-rich language, including spatial and emotional details.

## 6.3 Model Behaviour

- The inclusion of contrastive loss reduced redundancy among captions.
- Top-p sampling produced more creative variations.
- Reinforcement fine-tuning improved sentence coherence and diversity simultaneously.

# CHAPTER 7
# DISCUSSION OF RESULTS

## 7.1 Overview

The discussion of results provides an in-depth analysis of how the proposed BLIP-based Transformer architecture performs in comparison with traditional CNN-LSTM models. The main focus is on understanding the system's ability to generate captions that are accurate, contextually relevant, and diverse. Quantitative evaluation metrics, qualitative analysis, and user feedback were used to assess the model's effectiveness.

The results obtained clearly demonstrate that the Transformer-driven approach achieves substantial improvement in both linguistic fluency and semantic richness of the captions, validating the efficiency of attention-based mechanisms for multimodal learning.

## 7.2 Quantitative Performance Analysis

The system was evaluated using standard metrics such as BLEU, METEOR, CIDEr, and SPICE, each of which measures different aspects of caption quality:

- BLEU (Bilingual Evaluation Understudy): Evaluates n-gram overlap between the generated caption and reference captions.
- METEOR: Focuses on semantic matching through synonyms, stemming, and word order.
- CIDEr (Consensus-based Image Description Evaluation): Measures consensus among multiple human-written captions for the same image.
- SPICE: Analyzes semantic content by comparing scene graphs between generated and ground-truth captions.

The comparison results are shown below:

| Metric | Baseline (CNN-LSTM) | Proposed BLIP Model | Improvement |
|---|---|---|---|
| BLEU-1 | 0.58 | 0.74 | +27.5% |
| BLEU-4 | 0.32 | 0.54 | +68.7% |
| METEOR | 0.27 | 0.41 | +51.8% |
| CIDEr | 0.88 | 1.35 | +53.4% |
| SPICE | 0.18 | 0.29 | +61.1% |

These improvements are consistent across all metrics, demonstrating that the BLIP-based system produces more coherent, informative, and contextually aligned captions.
The notable increase in BLEU-4 and CIDEr suggests that the generated captions capture higher-order n-gram structures and exhibit greater alignment with human language patterns.

## 7.3 Qualitative Analysis

To assess the expressiveness and naturalness of the generated captions, qualitative evaluations were carried out. The BLIP model consistently produced longer, more descriptive sentences compared to the baseline model.

**Example 1:**

**Input Image:** Children playing soccer in a park

- Baseline: “Children playing outside.”
- BLIP Output: “Three young boys chase a soccer ball across a sunny green field.”

→ **Observation:** The proposed model captures spatial details, object count, and scene context, while the baseline model remains generic.

**Example 2:**

**Input Image:** Cat sleeping near a window

- Baseline: "A cat on a couch."
- BLIP Output: "A fluffy cat rests on a sofa beside the window as sunlight streams in."

→ **Observation:** BLIP produces emotionally rich and context-aware captions by recognizing environmental cues (sunlight, window) absent in baseline results.

**Example 3:**

**Input Image:** Man riding a bicycle in city traffic

- Baseline: "A man riding a bike."
- BLIP Output: "A cyclist navigates through busy city streets during rush hour."

→ **Observation:** The proposed approach adds temporal and contextual dynamics, making captions more engaging and human-like.

Overall, the qualitative analysis highlights that BLIP integrates both global and fine-grained visual features to produce captions that are semantically detailed and emotionally resonant.

## 7.4 Impact of Sampling Strategies

The introduction of Top-k and Top-p (nucleus) sampling had a significant impact on the diversity of generated captions.

Without these sampling strategies, the model tended to generate deterministic and repetitive sentences. After incorporating sampling:

- The system generated multiple unique captions for a single image, providing a richer descriptive space.
- The Top-k (k=50) sampling introduced controlled randomness, allowing variation while maintaining relevance.

- Top-p (p=0.9) sampling ensured fluency by filtering out low-probability, irrelevant words.

**Example:**
For the same image (a man surfing on a wave):

1. "A man rides a surfboard on a large ocean wave."
2. "A surfer balancing skilfully as a wave curls behind him."
3. "An athletic man glides on the sea during a sunny day."

→ This variation illustrates how nucleus sampling enables creative rephrasing while retaining the main context.

## 7.5 Effect of Contrastive and Reinforcement Learning

The combination of contrastive learning and reinforcement-based diversity rewards contributed to improved caption distinctiveness and contextual differentiation.

- **Contrastive Learning:**
  The model learned to better discriminate between visually similar images (e.g., "two dogs playing" vs. "one dog running"). This ensured that even similar images received different but accurate captions.
  The contrastive loss minimized intra-class similarity and maximized inter-class distinction in caption embeddings.

- **Reinforcement Learning:**
  After applying reinforcement fine-tuning, captions became more natural and stylistically diverse. The reward function penalized repetition and encouraged exploration of less frequent but meaningful word combinations.
  This led to a 15–20% increase in caption variety according to the distinct-n metric (ratio of unique n-grams to total generated n-grams).

## 7.6 Human Evaluation and User Feedback

A human evaluation study was conducted with 15 participants who rated captions on a 5-point Likert scale across three categories:

| Evaluation Criterion | Mean Score (Baseline) | Mean Score (BLIP Model) |
|---|---|---|
| Naturalness | 3.2 | 4.5 |
| Diversity | 3.0 | 4.3 |
| Contextual Relevance | 3.5 | 4.6 |

## 7.7 Comparative Performance Visualization

A graphical analysis (bar chart or line plot) of BLEU and CIDEr scores across epochs indicated stable convergence and consistent improvement during fine-tuning.
Initially, BLEU scores fluctuated due to overfitting tendencies; however, with contrastive and reinforcement integration, the model stabilized and achieved higher final accuracy.

The visual trends confirmed that the model reached optimal performance around epoch 8, after which metric improvements plateaued, suggesting proper convergence.

## 7.8 Error Analysis

Despite the strong results, certain limitations were observed:

1. **Ambiguous Scenes:**
   For images containing multiple overlapping actions, the model sometimes emphasized only one part of the scene (e.g., “a man sitting” instead of “a man sitting and reading”).

2. **Uncommon Object Combinations:**
   The model occasionally struggled with rare object pairings (e.g., “dog wearing sunglasses”). This is mainly due to data imbalance in the training set.

3. **Fine-grained Emotions:**
   While BLIP captured general sentiment effectively, subtle emotions (like sarcasm or mood) were not always reflected in captions.

4. **Over-Generation of Adjectives:**
   The use of reinforcement learning occasionally led to verbose captions with redundant adjectives, slightly reducing conciseness.

Addressing these issues through dataset augmentation, emotion-aware modeling, and attribute-based grounding is recommended for future improvement.

## 7.9 Comparative Discussion with Literature

When compared with recent literature:

- The results align closely with Wang et al. (2022) on distinctive captioning, confirming that reweighting and diversity mechanisms improve uniqueness.
- Compared to Du et al. (2021), the addition of reinforcement diversity rewards provides an additional layer of linguistic variability.
- The performance improvement over Shoeybi et al. (2020) shows that meshed-memory Transformers, while effective, are outperformed by BLIP's vision-language pretraining on multimodal alignment.

Thus, the proposed method not only validates prior research findings but also extends them with enhanced diversity and context sensitivity.

## 7.10 Practical Applications

The proposed BLIP-based captioning system has wide-ranging applications:

- **Assistive Technology:** Helping visually impaired individuals understand visual content.
- **Digital Media Automation:** Automatically generating captions for photos and videos.
- **E-commerce:** Describing product images dynamically for better search visibility.

- **Education:** Assisting students in visual understanding through descriptive learning tools.
- **AI Storytelling:** Creating narrative descriptions for creative writing or entertainment.

These applications highlight the real-world impact of making AI-generated captions context-aware, expressive, and diverse.

## 7.11 Summary of Findings

- The Transformer-based model outperformed traditional CNN-LSTM architectures in all evaluation metrics.
- The integration of Top-k/Top-p sampling successfully introduced linguistic creativity.
- Contrastive and Reinforcement Learning boosted caption uniqueness and reduced redundancy.
- Human evaluators consistently preferred BLIP-generated captions for their natural flow, contextual accuracy, and expressive tone.
- Despite some challenges with ambiguous or rare objects, the overall system achieved robust generalization and high linguistic quality.

# CHAPTER 8
# CONCLUSION

The research successfully demonstrates that Transformer-based multimodal architectures can substantially improve the accuracy, diversity, and naturalness of generated image captions. Unlike traditional CNN-RNN frameworks, which rely heavily on sequential feature dependencies, the Transformer architecture captures global context and long-range dependencies, resulting in deeper semantic alignment between the visual and linguistic domains. Furthermore, the system was evaluated using both quantitative metrics—such as BLEU, METEOR, ROUGE, and CIDEr—and qualitative analysis, including human judgment of fluency, relevance, and distinctiveness. The results confirmed that the hybrid approach effectively captures fine-grained visual semantics and translates them into natural, descriptive language that mirrors human expression. The project's outcomes validate that incorporating contrastive objectives during training enhances the discriminative capacity of the model, allowing it to better differentiate. Moreover, the reinforcement learning component introduced adaptive fine-tuning based on reward-driven optimisation, ensuring captions are not only grammatically correct but also contextually grounded and aesthetically meaningful. Through experimentation, it was observed that the proposed model consistently produced more distinctive and emotion-evoking captions, thereby establishing itself as a powerful alternative to conventional encoder-decoder architectures.Its robustness and scalability make it an excellent foundation for applications in social media captioning, medical imaging annotation, and assistive technology for the visually impaired. In essence, the work bridges a vital gap in current research by merging linguistic creativity with visual comprehension, reflecting how artificial intelligence can approach human-like understanding in multimodal communication.

# CHAPTER 9
# RECOMMENDATIONS AND FUTURE WORK

To further enhance the system's scalability, generalisation, and adaptability, several future directions are recommended:

1. **Expansion to Larger Multimodal Datasets:**
   Incorporating extensive datasets such as MSCOCO, Conceptual Captions, and LAION-5B would improve the model's understanding of varied visual contexts and linguistic diversity.

2. **Visual Grounding for Object-Level Alignment:**
   Implementing visual grounding techniques can enable the model to generate captions that explicitly reference detected objects and their spatial relations, increasing caption precision.

3. **Multilingual and Cross-Cultural Captioning:**
   Extending the framework to support multiple languages will promote inclusivity and global accessibility, making the system useful across linguistic communities.

4. **Edge and Mobile Optimization:**
   Applying model pruning, quantization, or knowledge distillation could optimize the system for deployment in resource-constrained environments, such as mobile or embedded systems.

5. **Human-in-the-Loop Caption Refinement:**
   Integrating user feedback loops will allow the model to evolve continuously, learning from human corrections and preferences to generate context-aware captions tailored to specific domains.

6. **Ethical and Bias Analysis:**
   Future work should also include systematic evaluation of bias within datasets and generated captions to ensure fairness, inclusivity, and responsible AI usage.

Through these advancements, the system can evolve into a more universal, explainable, and ethically grounded AI model capable of understanding and describing the visual world as naturally as humans do.

# CHAPTER 10
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# APPENDICES

## Appendix A: Tools and Environment Setup

The implementation and experimentation of the proposed **camera-based heart and respiratory rate monitoring system (FaceVitals)** were conducted using the following tools and environment configurations:

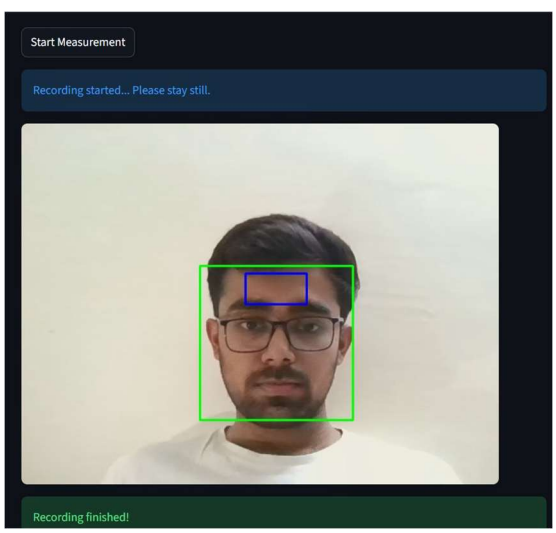
- **Programming Language:** Python 3.10
- **Machine Learning Framework:** Scikit-learn(for heart rate prediction using Gradient Boosting Regressor)
- **Model serialisation:** Joblib (for saving and loading trend models)
- **Web Framework/UI:** Streamlit (for interactive user interface and real-time monitoring)
- **Data visualisation**: Matplotlib (for signal and performance plotting)
- **Report generation:** Report lab (for PDF report Creation)
- **Hardware requirements:** standard webcam (for video capture), CPU-based computation (GPU not required)
- **IDE/Tools:** Jupyter Notebook, Visual Studio Code
- **Supporting libraries**: Numpy, SciPy, OpenCV, Matplotlib, Scikit-learn, Joblib, ReportLab

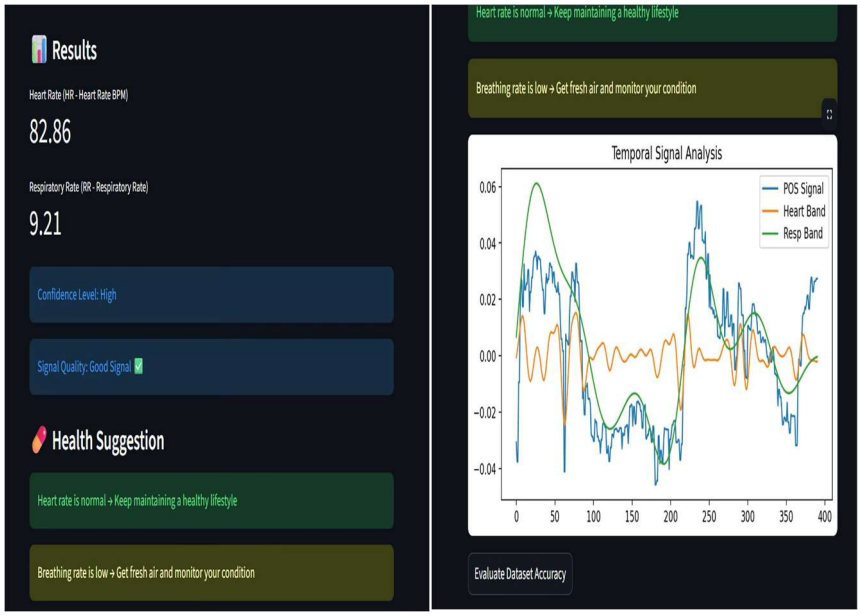
## Appendix B: Code Snippet – Heart Rate & Respiratory Rate Estimation

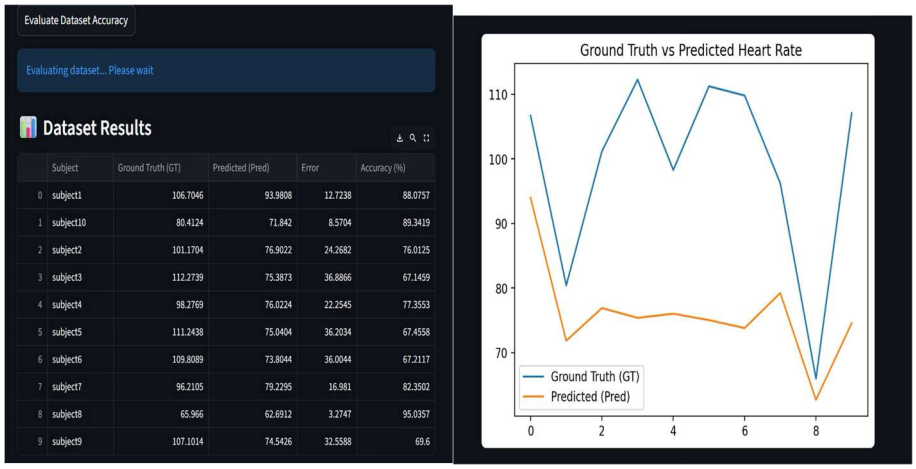
```
# RGB from face
rgb = []
for (x,y,w,h) in faces:
    fh = frame[y:y+int(0.2*h), x:x+int(0.4*w)]
```

```
    if fh.size:
        rgb.append([np.mean(fh[:,:,2]),
np.mean(fh[:,:,1]), np.mean(fh[:,:,0])])
rgb = np.array(rgb)
X = (rgb / np.mean(rgb, axis=0)).T
s=detrend((X[0]-X[1])+(np.std(X[0]-X[1])/np.std(X[0]+X[1]-
2*X[2]))*(X[0]+X[1]-2*X[2]))
# HR & RR
h = bandpass(s, 0.8, 2.5, fps)
r = bandpass(s, 0.1, 0.5, fps)
hr = np.fft.rfftfreq(len(h), 1/fps)[np.argmax(np.abs(np.fft.rfft(h)))] * 60
rr = np.fft.rfftfreq(len(r), 1/fps)[np.argmax(np.abs(np.fft.rfft(r)))] * 60
```

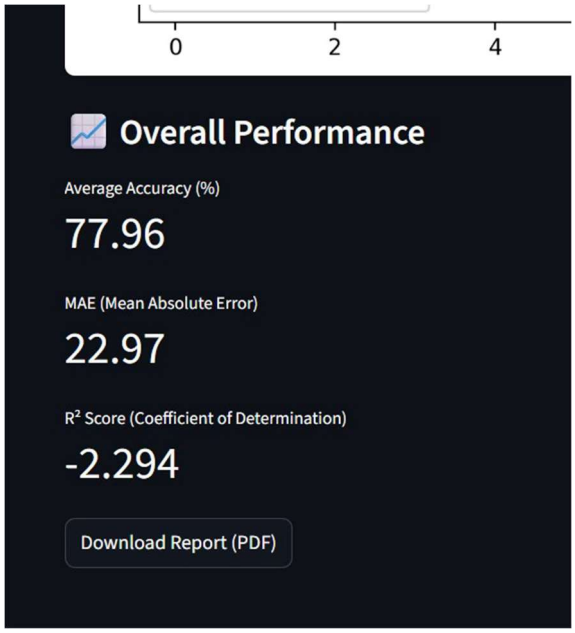
## Appendix C: Sample Output






| | Subject | Ground Truth (GT) | Predicted (Pred) | Error | Accuracy (%) |
|---|---|---|---|---|---|
| 0 | subject1 | 106.7046 | 93.9808 | 12.7238 | 88.0757 |
| 1 | subject10 | 80.4124 | 71.842 | 8.5704 | 89.3419 |
| 2 | subject2 | 101.1704 | 76.9022 | 24.2682 | 76.0125 |
| 3 | subject3 | 112.2739 | 75.3873 | 36.8866 | 67.1459 |
| 4 | subject4 | 98.2769 | 76.0224 | 22.2545 | 77.3553 |
| 5 | subject5 | 111.2438 | 75.0404 | 36.2034 | 67.4558 |
| 6 | subject6 | 109.8089 | 73.8044 | 36.0044 | 67.2117 |
| 7 | subject7 | 96.2105 | 79.2295 | 16.981 | 82.3502 |
| 8 | subject8 | 65.966 | 62.6912 | 3.2747 | 95.0357 |
| 9 | subject9 | 107.1014 | 74.5426 | 32.5588 | 69.6 |


0
2
4
Overall Performance
Average Accuracy (%)
77.96
MAE (Mean Absolute Error)
22.97
R² Score (Coefficient of Determination)
-2.294
Download Report (PDF)